

Class work problems

Unit IV: Engineering Optimization.

Formulation of LPP:

- 1) A calculator company produces a handheld calculator and a scientific calculator. Long-term projections indicate an expected demand of at least 150 scientific and 100 handheld calculators each day. Because of limitations on production capacity, no more than 250 scientific and 200 handheld calculators can be made daily. To satisfy a shipping contract, a minimum of 250 calculators must be shipped each day. If each scientific calculator sold, results in a 20 rupees loss, but each handheld calculator produces a 50 rupees profit; then how many of each type should be manufactured daily to maximize the net profit?
- 2) Vitamins B1 and B2 are found in two foods F1 and F2. 1 unit of F1 contains 3 units of B1 and 4 units of B2. 1 unit of F2 contains 5 units of B1 and 3 units of B2 respectively. Minimum daily prescribed consumption of B1 & B2 is 50 and 60 units respectively. Cost per unit of F1 & F2 is Rs. 6 & Rs. 3 respectively.

Formulation of Linear programming problem and solving by graphical method.

- 3) A firm is engaged in animal breeding. The animals are to be given nutrition supplements every day. There are two products A and B which contain the three required nutrients.

Nutrients	Quantity/unit		Minimum Requirement
	A	B	
1	72	12	216
2	6	24	72
3	40	20	200

Product cost per unit are: A: rs. 40; B: Rs. 80. Find out quantity of product A & B to be given to provide minimum nutritional requirement.

- 4) A firm makes two products P1 & P2 and has production capacity of 18 tonnes per day. P1 & P2 require same production capacity. The firm must supply at least 4 t of P1 & 6 t of P2 per day. Each tonne of P1 & P2 requires 60 hours of machine work each. Maximum machine hours available are 720. Profit per tonne for P1 is Rs. 160 & P2 is Rs. 240. Find optimal solution by graphical method.

Simplex method:

5)

Maximize:

$$Z = 3x_1 + 2x_2$$

Subject to:

$$x_1 + x_2 \leq 4$$

$$2x_1 + x_2 \leq 5$$

$$x_1, x_2 \geq 0$$

6)

Maximize:

$$Z = 3x_1 + 2x_2$$

Subject to:

$$x_1 + x_2 \leq 4$$

$$x_1 \leq 5$$

$$x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

Gradient decent method:

7. Find the minimum using gradient descent $f(x) = x^2 - 6x + 9$ from the point $x=0$, taking the learning rate 0.1. Perform 10 iterations.
8. Find the minimum using gradient descent $f(x) = e^{-x}$ from the point $x=0$, taking the learning rate 0.1. Perform 10 iterations.

Lagrange's multipliers:

9. Minimize the function $f(x, y) = x^2 + 2y^2$ subject to the constraint $x + y = 4$.
10. Minimize $f(x, y) = 2x + 3y$, subject to the constraint, $g(x, y) = x^2 + y^2 - 9 = 0$

KKT Conditions:

11. Minimize $f(x, y) = x^2 + y^2$, subject to the constraint $g(x, y) = x + y - 1 \leq 0$.
12. Maximize $f(x, y) = 3x + 4y$, subject to the constraint $g(x, y) = x^2 + y^2 - 4 \leq 0$.

Integer Linear Programming problem (Branch and Bound Method)

13.

$$\text{Maximize } Z = 5x + 4y$$

$$x + 2y \leq 8$$

$$3x + 2y \leq 10$$

$$x \geq 0, \quad y \geq 0$$

$$x_1, x_2 \in \mathbb{Z} \quad (\text{integers})$$

14. Solve the following problem by branch and bound method:

$$MAX Z = 100x_1 + 150x_2$$

subject to

$$8000x_1 + 4000x_2 \leq 40000$$

$$15x_1 + 30x_2 \leq 200$$

$$\text{and } x_1, x_2 \geq 0$$